Procedure for Resolving Critical Hardware/Environmental Faults on Huawei Access Node KR-ACC-02 at KR Puram Access Site

Node: KR-ACC-02 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_5832

Description

The Huawei Access Node KR-ACC-02, situated at the KR Puram Access site, is prone to critical hardware/environmental faults due to the inherent design of the ASIC memory parity error checking mechanism. As an Access node, it plays a pivotal role in the telecommunications network, facilitating the exchange of data packets between the core network and the customer premises equipment. The ASIC memory parity error checking mechanism is responsible for detecting and correcting errors that may occur during data transmission. However, in the event of a line card crash or a power supply redundancy loss, the node may experience a catastrophic failure, leading to a loss of connectivity and a subsequent ALARM_5832 alarm. This procedure is designed to help the network administrator diagnose and resolve such critical faults in a timely and efficient manner.

Pre-requisite Checks

- check: Verify that the remote hands are dispatched to the KR Puram Access site to facilitate physical access to the node in case of a hardware failure., description: Ensure that the remote hands are equipped with the necessary tools and expertise to perform the required tasks.

- check: Check the status of the redundant link to ensure that it is operational and can take over in case of a failure., description: Verify that the redundant link is properly configured and can be activated in case of a failure.

- check: Check the power supply redundancy to ensure that it is functioning properly., description: Verify that the power supply redundancy is operational and can take over in case of a failure.

Potential Root Causes

Cause	Probability
ASIC memory parity error	High
Line card crash	Medium
Power supply redundancy loss	Low

Diagnostic Commands & Expected Outputs

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Command: display fault, Expected Output: Fault Information: Fault Type: ASIC memory parity error Fault Location: KR-ACC-02 , Description: This command displays the fault information, including the fault type, location, and timestamp. The expected output indicates that the fault is an ASIC memory parity error, which is a critical
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fault that requires immediate attention.

Command: display fault history, Expected Output: Fault History: Fault Type: Line card crash Fault Location: KR-ACC-02 , Description: This command displays the fault history, including the fault type, location, and timestamp. The expected output indicates that the fault is a line card crash, which is a critical fault that requires immediate attention.

Command: display power supply, Expected Output: Power Supply Information: Power Supply Status: Redundant , Description: This command displays the power supply information, including the status and redundancy level. The expected output indicates that the power supply is redundant, which is a critical component that requires monitoring and maintenance.
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Resolution Steps

Step 1: Step: Replace the faulty line card, Command: replace line-card KR-ACC-02 1, Description: This command replaces the faulty line card with a new one. The new line card will be configured with the same settings as the previous one to ensure seamless connectivity.

Step 2: Step: Replace the faulty power supply, Command: replace power-supply KR-ACC-02 1, Description: This command replaces the faulty power supply with a new one. The new power supply will be configured with the same settings as the previous one to ensure seamless operation.

Step 3: Step: Verify the configuration, Command: display configuration, Description: This command displays the configuration of the node, including the settings and parameters. The output will be verified to ensure that the configuration is correct and consistent with the expected settings.

Post-Check Verification

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Command: display fault, Expected Output: Fault Information: Fault Type: None Fault Location: KR-ACC-02 , Description: This command verifies that the fault has been resolved by checking the fault information.

Command: display power supply, Expected Output: Power Supply Information: Power Supply Status: Redundant , Description: This command verifies that the power supply is functioning properly by checking the status and redundancy level.
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Roll-back Procedures

- Step: Restore the original configuration, Command: restore configuration KR-ACC-02 1, Description: This command restores the original configuration of the node, including the settings and parameters, in case of a failed fix.

- Step: Replace the faulty component with the original one, Command: replace line-card KR-ACC-02 1 original, Description: This command replaces the faulty component with the original one, in case of a failed fix.